

9913 Mark Lopes, BE COMPS A, NLP LAB 3

❖ Understanding Escape Characters and Raw Strings in Python

```
In [67]: print("\t this represents a tab") #escape characters are nongraphic, control  
        this represents a tab
```

```
In [68]: print("\\t this represents a tab")  
\\t this represents a tab
```

```
In [69]: print(r"\t this represents a tab") #r"xxxxxxx" .... raw string  
\\t this represents a tab
```

❖ Regular expression is very powerful method of matching pattern in text

A regular expression or RegEx is a special text string that helps to find patterns in data. A RegEx can be used to check if some pattern exists in a string/document/ corpus. To use RegEx in python first we should import the RegEx module which is called re.

```
In [70]: import re
```

Regular Expression Methods in Python (`re` module)

To find patterns in text, Python's `re` module provides several useful methods that allow you to search, extract, split, or replace based on regular expressions.

`re.match()`: searches only in the beginning of the first line of the string and returns matched objects if found, else returns None.

`re.search`: Returns a match object if there is one anywhere in the string, including multiline strings.

`re.findall`: Returns a list containing all matches `re.split`: Takes a string, splits it at

the match points, returns a list `re.sub`: Replaces one or many matches within a

string

❖ `re.match(pattern, string)`

- Description: Searches for a pattern only at the beginning of the string.
- Returns: A match object if found at the start; otherwise, returns `None` .

```
re.match(r'\d+', '123abc') # ❖ Match found: '123'
```

```
re.match(r'\d+', 'abc123') # ❖ No match (does not start with digits)
```

Returns: A match object **if** found; otherwise, returns **None**.

In [70]:

❖ `re.search(pattern, string)`

Description:

Searches for the first occurrence of the pattern anywhere in the string.

Returns:

A match object if found; otherwise, returns `None`.

```
re.search(r'\d+', 'abc123') # ❖ Match found: '123'
```

❖ `re.search(pattern, string)`

`re.findall(pattern, string)`

Description:

Finds all non-overlapping matches of the pattern in the string.

Returns:

A list of matched substrings.

```
re.findall(r'\d+', 'abc123def456') # ❖ Returns: ['123', '456']
```

✂ `re.split(pattern, string)`

Description:

Splits the string at each match of the pattern.

Returns:

A list of substrings.

```
import re
```

```
re.split(r'\s+', 'This is\ta test') # ❖ Returns: ['This', 'is',  
'a', 'test'] #is a form of basic word  
tokenization.
```

❖ `re.sub(pattern, replacement, string)`

Description:

Replaces one or more matches of the pattern in the string with a replacement.

Returns:

A new string with replacements applied.

```
import re
```

```
re.sub(r'\d+', '#', 'Order 123 is now 456') # Returns: 'Order # is now #'
```

Examples: *Match*

```
In [71]: txt = 'I love to teach python and javaScript'
# It returns an object with span, and match
match = re.match('I love to teach', txt, re.I) #re.I means ignore case
print(match)
# <re.Match object; span=(0, 15), match='I love to teach'>
# We can get the starting and ending position of the match as tuple using span()

span = match.span()
print(span) # (0, 15)

# Lets find the start and stop position from the
span start, end = span print(start, end) # 0, 15

substring = txt[start:end]
print(substring) # I love to teach

<re.Match object; span=(0, 15), match='I love to teach'>
(0, 15)
0 15
I love to teach
```

```
In [72]: import re txt = 'I love to teach python and
javaScript' match = re.match('javaScript',
txt, re.I) print(match) # None
```

None

Examples : Search

```
In [73]: txt = '''Python is the most beautiful language that a human being has ever
cre
I recommend python for a first programming language'''

# It returns an object with span and match
match = re.search('first', txt, re.I)
print(match) # <re.Match object; span=(100, 105), match='first'>
# We can get the starting and ending position of the match as tuple using
span span = match.span() print(span)      # (100, 105)
# Lets find the start and stop position from the
span start, end = span print(start, end) # 100 105
substring = txt[start:end] print(substring) #
first

<re.Match object; span=(100, 105), match='first'>
(100, 105)
100 105
first
```

Searching for All Matches Using findall

```
In [74]: txt = '''Python is the most beautiful language that a human being has ever
cre
I recommend python for a first programming language'''

# It return a list matches =
re.findall('language', txt, re.I)
print(matches) # ['language', 'language']
['language', 'language']
```

```
In [75]: txt = '''Python is the most beautiful language that a human being has ever
cre
I recommend python for a first programming language'''

# It returns list matches =
re.findall('python', txt, re.I)
print(matches) # ['Python', 'python']
['Python', 'python']
```

Since we are using re.I both lowercase and uppercase letters are included. If we do not have the re.I flag, then we will have to write our pattern differently.

```
In [76]: import re
txt = '''Python is the most beautiful language that a human being has ever
cre
I recommend python for a first programming language'''

matches = re.findall('Python|python', txt)
print(matches) # ['Python', 'python']
```

```
#
matches = re.findall('[Pp]ython', txt) print(matches) #
['Python', 'python']
```

```
['Python', 'python']
['Python', 'python']
```

Replacing a Substring

```
In [77]: txt = '''Python is the most beautiful language that a human being has ever
cre I recommend python for a first programming language'''

match_replaced = re.sub('Python|python', 'JavaScript', txt)
print(match_replaced) # JavaScript is the most beautiful language that a huma
# OR
match_replaced = re.sub('[Pp]ython', 'JavaScript', txt)
print(match_replaced) # JavaScript is the most beautiful language that a huma
```

JavaScript is the most beautiful language that a human being has ever created.

I recommend JavaScript for a first programming language JavaScript is the most beautiful language that a human being has ever created.

I recommend JavaScript for a first programming language

The following string is really hard to read unless we remove the % symbol. Replacing the % with an empty string will clean the text.

```
In [78]: txt = '''%I a%m te%%a%che%r% a%n%d %% I l%o%ve te%ach%ing.
T%he%re i%s n%o%th%ing as r%ewarding a%s e%duc%at%i%ng a%n%d e%m%p%ow%er%ing
p I fo%und te%a%ch%ing m%ore i%n%t%er%%e%st%ing t%h%an any o%ther %jobs. D%oes
this m%ot%iv%a%te %y%o%u to b%e a t%e%a%ch%er?'''

matches = re.sub('%', '', txt)
print(matches)
```

I am teacher and I love teaching.

There is nothing as rewarding as educating and empowering people.

I found teaching more interesting than any other jobs.

Does this motivate you to be a teacher?

Splitting Text Using RegEx Split

```
In [79]: txt = '''I am teacher and I love teaching.
There is nothing as rewarding as educating and empowering people.
I found teaching more interesting than any other jobs.
Does this motivate you to be a teacher?'''

print(re.split('\n', txt)) # splitting using \n - end of line symbol
```

```
['I am teacher and I love teaching.', 'There is nothing as rewarding as
educat ing and empowering people.', 'I found teaching more interesting than
any other
jobs.', 'Does this motivate you to be a teacher?']
```

Writing RegEx Patterns

```
In [80]: import re

regex_pattern = r'apple'
txt = 'Apple and banana are fruits. An old cliché says an apple a day a doctor'
matches = re.findall(regex_pattern, txt) print(matches) # ['apple']

# To make case insensitive adding flag '
matches = re.findall(regex_pattern, txt,
re.I) print(matches) # ['Apple', 'apple'] #
or we can use a set of characters method
regex_pattern = r'[Aa]pple' # this mean the first letter could be Apple or
ap matches = re.findall(regex_pattern, txt) print(matches) # ['Apple',
'apple']

['apple']
['Apple', 'apple']
['Apple', 'apple']
```

Square Bracket

```
In [81]: regex_pattern = r'[Aa]pple' # this square bracket mean either A or a txt =
'Apple and banana are fruits. An old cliché says an apple a day a doctor'
matches = re.findall(regex_pattern, txt) print(matches) # ['Apple', 'apple']

['Apple', 'apple']

In [82]: regex_pattern = r'[Aa]pple|[Bb]anana' # this square bracket means either A
or txt = 'Apple and banana are fruits. An old cliché says an apple a day a
doctor' matches = re.findall(regex_pattern, txt)
print(matches) # ['Apple', 'banana', 'apple', 'banana']

['Apple', 'banana', 'apple', 'banana']

In [83]: regex_pattern = r'\d' # \d is a special character which means digits txt =
'This regular expression example was made on December 6, 2019 and revis
matches = re.findall(regex_pattern, txt)
print(matches) # ['6', '2', '0', '1', '9', '8', '2', '0', '2', '1'], this is

['6', '2', '0', '1', '9', '8', '2', '0', '2', '1']

In [84]: regex_pattern = r'\d+' # \d is a special character which means digits, +
mean txt = 'This regular expression example was made on December 6, 2019
and revis matches = re.findall(regex_pattern, txt) print(matches) # ['6',
'2019', '8', '2021'] - now, this is better!

['6', '2019', '8', '2021']
```

```
In [85]: matches = re.findall( r'a.', "Apple and banana are fruits")
print(matches) # ['an', 'an', 'an', 'a ', 'ar']

matches = re.findall(r'a.+', "Apple and banana are fruits") # . any
character, print(matches) # ['and banana are fruits']

['an', 'an', 'an', 'a ', 'ar']
['and banana are fruits']
```

```
In [86]: txt = '''I am not sure if there is a convention how to write the word e-
mail. Some people write it as email others may write it as Email or E-
mail.''' regex_pattern = r'[Ee]-?mail' # ? means here that '-' is optional
matches = re.findall(regex_pattern, txt)
print(matches) # ['e-mail', 'email', 'Email', 'E-mail']

['e-mail', 'email', 'Email', 'E-mail']
```

```
In [87]: txt = '''This regular expression example was made on December 6, 2019 and
rev This is This.''' regex_pattern = r'^This' # ^ means starts with matches
= re.findall(regex_pattern, txt, re.MULTILINE) print(matches) # ['This']

['This', 'This']
```

```
In [88]: # Example text with multiple lines
import re # Import the regular expressions module

# Multiline text input text
= """Hello, how are you?
Hello World!
Hi there,
Hello again!"""

pattern = r'^Hello'
# Regular expression pattern
# ^Hello matches lines that start with 'Hello'
# ^ asserts start of a line (because of re.MULTILINE)

# Split the text into individual lines
lines = text.splitlines()

# Print the matches along with their line numbers
print("Matches found on lines:")

# Loop over each line with its line number (starting from 1)
for i, line in enumerate(lines, start=1):
    # Use re.match to check if the line starts with 'Hello'
    if re.match(pattern, line):
        # Print the matching line along with its line number
        print(f"Line {i}: {line}")
```

Matches found on lines:

Line 1: Hello, how are you?

Line 2: Hello World!

```
txt = 'This regular expression example was made on December 6, 2019 and  
revis regex_pattern = r'^[A-Za-z ]+' # ^ in set character means negation,  
not A to matches = re.findall(regex_pattern, txt) print(matches) # ['6,',  
'2019', '8,', '2021']
```

Line 4: Hello again!

In [89]:

```
['6,', '2019', '8,', '2021']
```

In [90]: **import** re # Import the regular expression module

```
# Define a text string containing an email address  
statement = 'Please contact us at:  
sup.por.t@datacamp.com'  
  
# Use re.search() to find the first email pattern in the  
string # The pattern explanation:  
# ([\w.-]+) -> Group 1: username part (letters, digits, underscore, dot, or  
h  
# @ -> Matches the @ symbol literally  
# ([\w.-]+) -> Group 2: domain/host part (same character set)  
# r'' -> Raw string so that backslashes are interpreted correctly  
match = re.search(r'([\w.-]+)@([\w.-]+)', statement) #square bracket inside  
do  
  
# Check if a match was found (corrected from if statement: to if match:)  
if match:  
    # Print the full matched email address (entire match)  
    print("Email address:", match.group())  
  
    # Print the first capturing group: the username part of the email  
    print("Username:", match.group(1))  
  
    # Print the second capturing group: the host/domain part of the email  
    print("Host:", match.group(2))  
else:  
    print("No email address found.")
```

Email address: sup.por.t@datacamp.com

Username: sup.por.t

Host: datacamp.com

In [91]: `import re`

```
# Input string
string = "From : using the : character"

# re.findall will search for all substrings that:
# - Start with 'F' at the beginning of the string (^F)
# - Followed by the shortest possible sequence of any characters (.+?)
# Non-greedy match ensures it stops at the first point it
can. x = re.findall(r"^F.+?", string)

# Print the matched substring
print(x)
```

`['Fr']`

In [91]:

In [92]: `exampleString = '''Jessica is 15 years old, and Daniel is 27 years old. Edward is 97 years old, and his grandfather, Oscar, is 102.`

```
''' ages = re.findall(r'\d{1,3}', exampleString)
#names = re.findall(r'[A-Z][a-
z]*', exampleString) names =
re.findall(r'\w*', exampleString)
```

```
print(ages)
print(names)
```

`['15', '27', '97', '102']`
`['Jessica', '', 'is', '', '15', '', 'years', '', 'old', '', '', 'and', '', 'Dan iel', '', 'is', '', '27', '', 'years', '', 'old', '', '', 'Edward', '', 'is', '', '97', '', 'years', '', 'old', '', '', 'and', '', 'his', '', 'grandfather', '', '', 'Oscar', '', '', 'is', '', '102', '', '', '']`

In [93]: `txt = 'This regular expression example was made on December 6, 2019 and revis regex_pattern = r'\d{4}' # exactly four times matches = re.findall(regex_pattern, txt) print(matches) # ['2019', '2021']`

```
txt = 'This regular expression example was made on December 6, 2019 and revis regex_pattern = r'\d{1,4}' # 1 to 4 matches = re.findall(regex_pattern, txt) print(matches) # ['6', '2019', '8', '2021']
```

`['2019', '2021']`

`['6', '2019', '8', '2021']`

```
In [94]: import re pattern = r"\bcat\b" # Match 'cat' as a complete word (with word
        boundaries)

        # 1st Match match = re.search(pattern,
        "The cat sat!") if match: print('Match
        1')

        # 2nd Match match = re.search(pattern, "We
        s>cat<tered?") if match: print('Match 2')

        # 3rd Match match = re.match(pattern,
        "We scattered") if match:
            print('Match 3')
```

Match 1

Match 2

🔍 Task: Complete the Given Code Using Your Understanding of Regular Expressions

You are provided with partial code for four questions. Your task is to complete each one by replacing the `####` placeholders with the appropriate logic or function calls. Make sure to test your solutions and validate the output.

Q.1 What is the most frequent word in the following paragraph?

paragraph = 'I love teaching. If you do not love teaching what else can you love. I love Python if you do not love something which can give you all the capabilities to develop an application what else can you love.'

Q.2 The position of some particles on the horizontal x-axis are -12, -4, -3 and -1 in the negative direction, 0 at origin, 4 and 8 in the positive direction. Extract these numbers from this whole text and find the distance between the two furthest particles points = ['-12', '-4', '-3', '-1', '0', '4', '8']
sorted_points = [-12, -4, -3, -1, -1, 0, 2, 4, 8] distance = 8 - (-12) # 20

Q.3 Write a pattern which identifies if a string is a valid python variable

is_valid_variable('first_name') # True is_valid_variable('first-name') # False
is_valid_variable('1first_name') # False is_valid_variable('firstname') # True

Q.4 Clean the following text. After cleaning, count three most frequent words in the string.

sentence = "%I \$am@% a %tea@cher%, &and& I lo%#ve %tea@ching%;. There \$is nothing; &as& mo@re rewarding as educa@ting &and& @emp%o@wering peo@ple. ;I found tea@ching m%o@re interesting tha@n any other %jo@bs."

```
%Do@es thi%s mo@tivate yo@u to be a tea@cher!?" print(clean_text(sentence));
```

I am a teacher and I love teaching There is nothing as more rewarding as educating and empowering people I found teaching more interesting than any other jobs Does this motivate you to be a teacher

```
print(most_frequent_words(cleaned_text)) # [(3, 'I'), (2, 'teaching'), (2, 'teacher')]
```

Q.1 What is the most frequent word in the following paragraph?

paragraph = 'I love teaching. If you do not love teaching what else can you love. I love Python if you do not love something which can give you all the capabilities to develop an application what else can you love.'

```
In [95]: import re
from collections import Counter # Import Counter class for counting frequency

paragraph = '''I love teaching. If you do not love teaching what else can
you I love Python if you do not love something which can give you all the
capabili to develop an application what else can you love.'''

# Step 1: Convert to lowercase
paragraph = paragraph.lower()

# Step 2: Remove punctuation
paragraph = re.sub(r'^\w\s', '', paragraph)

# Step 3: Split into words
# Hint: Use split() to break the paragraph into a list of words
words = paragraph.split()

# Step 4: Count frequency
# Hint: Use Counter to count frequency of each word in the list
word_counts = Counter(words)

# Step 5: Find most frequent word
# Hint: Use most_common(1)[0] to get the top word and its
count most_common = word_counts.most_common(1)[0]
print(most_common)

('love', 6)
```

Q.2 The position of some particles on the horizontal x-axis are -12, -4, -3 and -1 in the negative direction, 0 at origin, 4 and 8 in the positive direction. Extract these numbers from this whole text and find the distance between the two furthest particles points = ['-12', '-4', '-3', '-1', '0', '4', '8'] sorted_points = [-12, -4, -3, -1, -1, 0, 2, 4, 8] distance = 8 - (-12) # 20

```
In [96]: import re # Import regular expressions module

text = '''The position of some particles on the horizontal x-axis are -12, -4,
0 at origin, 4 and 8 in the positive direction.'''

# Step 1: Extract all integers including negatives
points = re.findall(r'[-]?\d+', text)
print("Extracted points:", points)

# Step 2: Convert to integers
points = [int(p) for p in points] # Hint: use int(p) to convert each string
t

# Step 3: Sort and find distance
min_point = min(points) # Hint: use min(points) to get the small
max_point = max(points) # Hint: use max(points) to get the larg distance
distance = max_point - min_point # Hint: subtract min_point from max_po
print("Distance between furthest particles:", distance)
```

Extracted points: ['-12', '-4', '-3', '-1', '0', '4', '8']

Distance between furthest particles: 20

Q.3 Write a pattern which identifies if a string is a valid python variable is_valid_variable('first_name') # True is_valid_variable('first-name') # False is_valid_variable('1first_name') # False is_valid_variable('firstname') # True

```
In [97]: def is_valid_variable(name):
# Python variable naming rules:
# - Must start with a letter (a-z or A-Z) or underscore (_)
# - Can be followed by letters, digits (0-9), or underscores
# Hint: use \w for "word character" (letter, digit, underscore), and
combi
# Hint: for first character, allow [a-zA-Z_], and for rest use \w*
pattern = r'^[a-zA-Z_]\w*$'
return re.match(pattern, name) is not None

# Test Cases
print(is_valid_variable('first_name')) # True
print(is_valid_variable('first-name')) #
False print(is_valid_variable('1first_name'))
# False
print(is_valid_variable('firstname')) # True
```

True

False

False

True

Q.4 Clean the following text. After cleaning, count the three most frequent words in the string.

```

sentence = '''%I am@ is nothing; &as& mo@re rewarding as educa@ting
&and& @emp%o@wering peo@ple.
;I found tea@ching m%o@re interesting tha@n any other %jo@bs.
%Do@es thi@s mo@tivate yo@u to be a tea@cher!?''''

```

```

print(clean_text(sentence))

```

```

# Expected cleaned output:
# I am a teacher and I love teaching There is nothing as more
rewarding as educating and empowering people
# I found teaching more interesting than any other jobs Does this
motivate you to be a teacher

```

```

print(most_frequent_words(cleaned_text))
# Expected output:
# [('I', 3), ('teaching', 2), ('teacher', 2)]

```

```

In [98]: ### Q.4 Clean the following text. After cleaning, count the three most
frequent def clean_text(text): text = re.sub(r'^a-zA-Z\s]', '', text)
#remove non alphabetic chars text = re.sub(r'\s+', ' ', text).strip() #if
more than one space character be

sentences = re.split(r'(?<=[.?!])\s*',text) #split into list of sentences
ba cleaned_sentences = [s.strip().capitalize() for s in sentences if s]
return " ".join(cleaned_sentences)

def most_frequent_words(text, n=3):
    words =
    text.lower().split()
    counts=Counter(words)
    return
    counts.most_common(n)

sentence = '''%I am@ is nothing; &as& mo@re rewarding as educa@ting &and&
@em ;I found tea@ching m%o@re interesting tha@n any other %jo@bs.
%Do@es thi@s mo@tivate yo@u to be a tea@cher!?''''

cleaned_text = clean_text(sentence)
print(cleaned_text)

# Expected cleaned output:
# I am a teacher and I love teaching There is nothing as more rewarding as
edu # I found teaching more interesting than any other jobs Does this motivate
you

I print(most_frequent_words(cleaned_text))
am # Expected output:
is # [('I', 3), ('teaching', 2), ('teacher', 2)]

nothing as more rewarding as educating and empowering people i found te aching
more interesting than any other jobs does this motivate you to be a teac

```

```
her [('i', 2), ('as', 2),  
('more', 2)]
```

POSTLAB

🔗 TASK A:

Refer to the provided link and complete the task as described.

The required text file has been uploaded in the classroom under the corresponding experiment.

<https://www.py4e.com/html3/11-regex?authuser=0>:

```
In [99]: # Load file once with open("/content/nlp-  
lab3.txt", "r") as f: lines = f.readlines()  
  
# Now `lines` is a list of all lines in the file.
```

```
In [100... # Search for lines that contain 'From'  
for i, line in enumerate(lines):  
    if "from" in line.lower():  
        print(i, line.strip()) #print line number and line
```

0 From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008
2 Received: from murder (mail.umich.edu [141.211.14.90])
6 Received: from murder ([unix socket])
9 Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.14.79])
12 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk [194.35.219.184])
15 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
21 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
25 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu [134.68.220.122]) 28 Received: from nakamura.uits.iupui.edu (localhost [127.0.0.1])
31 Received: (from apache@localhost)
37 From: stephen.marquard@uct.ac.za
57 SAK-12501 merge to 2-5-x: r39622, r39624:5, r39632:3 (resolve conflict from differing linebreaks for r39622) 60 This automatic notification message was sent by Sakai Collab (<https://collab.sakaiproject.org/portal>) from the Source site.
65 From louis@media.berkeley.edu Fri Jan 4 18:10:48 2008
67 Received: from murder (mail.umich.edu [141.211.14.97])
71 Received: from murder ([unix socket])
74 Received: from icestorm.mr.itd.umich.edu (icestorm.mr.itd.umich.edu [141.211.93.149]) 77 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk [194.35.219.184])
80 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
86 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
90 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu [134.68.220.122]) 93 Received: from nakamura.uits.iupui.edu (localhost [127.0.0.1])
96 Received: (from apache@localhost)
102 From: louis@media.berkeley.edu
125 This automatic notification message was sent by Sakai Collab (<https://collab.sakaiproject.org/portal>) from the Source site.
130 From zqian@umich.edu Fri Jan 4 16:10:39 2008
132 Received: from murder (mail.umich.edu [141.211.14.25])
136 Received: from murder ([unix socket])
139 Received: from ghostbusters.mr.itd.umich.edu (ghostbusters.mr.itd.umich.edu [141.211.93.144])
142 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk [194.35.219.184])
145 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
151 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
155 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu [134.68.220.122]) 158 Received: from nakamura.uits.iupui.edu (localhost [127.0.0.1])
161 Received: (from apache@localhost)

167 From: zqian@umich.edu
189 This automatic notification message was sent by Sakai Collab
(https://collab
b.sakaiproject.org/portal) from the Source site.
194 From rjlowe@iupui.edu Fri Jan 4 15:46:24
2008
196 Received: from murder (mail.umich.edu [141.211.14.25])
200 Received: from murder ([unix socket])
203 Received: from dreamcatcher.mr.itd.umich.edu
(dreamcatcher.mr.itd.umich.edu [141.211.14.43])
206 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
209 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
215 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
219 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 222 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
225 Received: (from apache@localhost)
231 From: rjlowe@iupui.edu
257 This automatic notification message was sent by Sakai Collab
(https://collab
b.sakaiproject.org/portal) from the Source site.
262 From zqian@umich.edu Fri Jan 4 15:03:18 2008
264 Received: from murder (mail.umich.edu [141.211.14.46])
268 Received: from murder ([unix socket])
271 Received: from firestarter.mr.itd.umich.edu (firestarter.mr.itd.umich.edu
[141.211.14.83])
274 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
277 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
283 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
287 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 290 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
293 Received: (from apache@localhost)
299 From: zqian@umich.edu
330 This automatic notification message was sent by Sakai Collab
(https://collab
b.sakaiproject.org/portal) from the Source site.
335 From rjlowe@iupui.edu Fri Jan 4 14:50:18
2008
337 Received: from murder (mail.umich.edu [141.211.14.93])
341 Received: from murder ([unix socket])
344 Received: from eyewitness.mr.itd.umich.edu (eyewitness.mr.itd.umich.edu
[14
1.211.93.142]) 347 Received: FROM paploo.uhi.ac.uk
(appl.prod.collab.uhi.ac.uk [194.35.219.18
4])
350 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])

356 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
360 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 363 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
366 Received: (from apache@localhost)
372 From: rjlowe@iupui.edu
409 This automatic notification message was sent by Sakai Collab
(https://colla
b.sakaiproject.org/portal) from the Source site.
414 From cwen@iupui.edu Fri Jan 4 11:37:30 2008
416 Received: from murder (mail.umich.edu [141.211.14.46])
420 Received: from murder ([unix socket])
423 Received: from tadpole.mr.itd.umich.edu (tadpole.mr.itd.umich.edu [141.21
1.14.72]) 426 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
429 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
435 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
439 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 442 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
445 Received: (from apache@localhost)
451 From: cwen@iupui.edu
488 This automatic notification message was sent by Sakai Collab
(https://colla
b.sakaiproject.org/portal) from the Source site.
493 From cwen@iupui.edu Fri Jan 4 11:35:08 2008
495 Received: from murder (mail.umich.edu [141.211.14.46])
499 Received: from murder ([unix socket])
502 Received: from it.mr.itd.umich.edu (it.mr.itd.umich.edu [141.211.93.151])
505 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
508 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
514 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
518 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 521 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
524 Received: (from apache@localhost)
530 From: cwen@iupui.edu
567 This automatic notification message was sent by Sakai Collab
(https://colla
b.sakaiproject.org/portal) from the Source site.
572 From gsilver@umich.edu Fri Jan 4 11:12:37 2008
574 Received: from murder (mail.umich.edu [141.211.14.25])
578 Received: from murder ([unix socket])
581 Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu
[141.211.14.7
9])

584 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
587 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
593 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
597 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 600 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
603 Received: (from apache@localhost)
609 From: gsilver@umich.edu
633 This automatic notification message was sent by Sakai Collab
(https://colla
b.sakaiproject.org/portal) from the Source site.
638 From gsilver@umich.edu Fri Jan 4 11:11:52 2008
640 Received: from murder (mail.umich.edu [141.211.14.36])
644 Received: from murder ([unix socket])
647 Received: from creepshow.mr.itd.umich.edu (creepshow.mr.itd.umich.edu [14
1.211.14.84]) 650 Received: FROM paploo.uhi.ac.uk
(appl.prod.collab.uhi.ac.uk [194.35.219.18
4])
653 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
659 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
663 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 666 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
669 Received: (from apache@localhost)
675 From: gsilver@umich.edu
699 This automatic notification message was sent by Sakai Collab
(https://colla
b.sakaiproject.org/portal) from the Source site.
704 From zqian@umich.edu Fri Jan 4 11:11:03 2008
706 Received: from murder (mail.umich.edu
[141.211.14.97])
710 Received: from murder ([unix socket])
713 Received: from carrie.mr.itd.umich.edu (carrie.mr.itd.umich.edu
[141.211.9
3.152]) 716 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
719 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
725 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
729 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 732 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
735 Received: (from apache@localhost)
741 From: zqian@umich.edu
764 This automatic notification message was sent by Sakai Collab
(https://colla
b.sakaiproject.org/portal) from the Source site.
769 From gsilver@umich.edu Fri Jan 4 11:10:22 2008

771 Received: from murder (mail.umich.edu [141.211.14.39])
775 Received: from murder ([unix socket])
778 Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu
[141.211.14.7
9])
781 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
784 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
790 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
794 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 797 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
800 Received: (from apache@localhost)
806 From: gsilver@umich.edu
830 This automatic notification message was sent by Sakai Collab
(https://colla
b.sakaiproject.org/portal) from the Source site.
835 From wagnermr@iupui.edu Fri Jan 4 10:38:42 2008
837 Received: from murder (mail.umich.edu [141.211.14.90])
841 Received: from murder ([unix socket])
844 Received: from shining.mr.itd.umich.edu (shining.mr.itd.umich.edu [141.21
1.93.153]) 847 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
850 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
856 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
860 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 863 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
866 Received: (from apache@localhost)
872 From: wagnermr@iupui.edu
899 This automatic notification message was sent by Sakai Collab
(https://colla
b.sakaiproject.org/portal) from the Source site.
904 From zqian@umich.edu Fri Jan 4 10:17:43 2008
906 Received: from murder (mail.umich.edu [141.211.14.97])
910 Received: from murder ([unix socket])
913 Received: from creepshow.mr.itd.umich.edu (creepshow.mr.itd.umich.edu [14
1.211.14.84]) 916 Received: FROM paploo.uhi.ac.uk
(appl.prod.collab.uhi.ac.uk [194.35.219.18
4])
919 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
925 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
929 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 932 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
935 Received: (from apache@localhost)
941 From: zqian@umich.edu

964 This automatic notification message was sent by Sakai Collab
(https://collab.b.sakaiproject.org/portal) from the Source site.
969 From antranig@caret.cam.ac.uk Fri Jan 4 10:04:14 2008
971 Received: from murder (mail.umich.edu [141.211.14.25])
975 Received: from murder ([unix socket])
978 Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu
[141.211.14.7
9])
981 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
984 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
990 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
994 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 997 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1000 Received: (from apache@localhost)
1006 From: antranig@caret.cam.ac.uk
1041 This automatic notification message was sent by Sakai Collab
(https://collab.sakaiproject.org/portal) from the Source site.
1046 From gopal.ramasammycook@gmail.com Fri Jan 4 09:05:31 2008
1048 Received: from murder (mail.umich.edu [141.211.14.90])
1052 Received: from murder ([unix socket])
1055 Received: from guys.mr.itd.umich.edu (guys.mr.itd.umich.edu
[141.211.14.7
6])
1058 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
1061 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1067 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
1071 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1074 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1077 Received: (from apache@localhost)
1083 From: gopal.ramasammycook@gmail.com
1115 This automatic notification message was sent by Sakai Collab
(https://collab.sakaiproject.org/portal) from the Source site.
1120 From david.horwitz@uct.ac.za Fri Jan 4 07:02:32 2008
1122 Received: from murder (mail.umich.edu [141.211.14.39])
1126 Received: from murder ([unix socket])
1129 Received: from guys.mr.itd.umich.edu (guys.mr.itd.umich.edu
[141.211.14.7
6])
1132 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
1135 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1141 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])

1145 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1148 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1151 Received: (from apache@localhost)
1157 From: david.horwitz@uct.ac.za
1201 This automatic notification message was sent by Sakai Collab
(<https://collab.sakaiproject.org/portal>) from the Source site.
1206 From david.horwitz@uct.ac.za Fri Jan 4 06:08:27 2008
1208 Received: from murder (mail.umich.edu [141.211.14.98])
1212 Received: from murder ([unix socket])
1215 Received: from firestarter.mr.itd.umich.edu
(firestarter.mr.itd.umich.edu
[141.211.14.83])
1218 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
1221 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1227 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
1231 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1234 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1237 Received: (from apache@localhost)
1243 From: david.horwitz@uct.ac.za
1270 This automatic notification message was sent by Sakai Collab
(<https://collab.sakaiproject.org/portal>) from the Source site.
1275 From david.horwitz@uct.ac.za Fri Jan 4 04:49:08 2008
1277 Received: from murder (mail.umich.edu [141.211.14.92])
1281 Received: from murder ([unix socket])
1284 Received: from galaxyquest.mr.itd.umich.edu
(galaxyquest.mr.itd.umich.edu
[141.211.93.145])
1287 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
1290 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1296 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
1300 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1303 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1306 Received: (from apache@localhost)
1312 From: david.horwitz@uct.ac.za
1348 This automatic notification message was sent by Sakai Collab
(<https://collab.sakaiproject.org/portal>) from the Source site.
1353 From david.horwitz@uct.ac.za Fri Jan 4 04:33:44 2008
1355 Received: from murder (mail.umich.edu [141.211.14.46])
1359 Received: from murder ([unix socket])
1362 Received: from workinggirl.mr.itd.umich.edu
(workinggirl.mr.itd.umich.edu
[141.211.93.143])

1365 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
1368 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1374 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
1378 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1381 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1384 Received: (from apache@localhost)
1390 From: david.horwitz@uct.ac.za
1434 This automatic notification message was sent by Sakai Collab
(https://collab.sakaiproject.org/portal) from the Source site.
1439 From stephen.marquard@uct.ac.za Fri Jan 4 04:07:34 2008
1441 Received: from murder (mail.umich.edu [141.211.14.25])
1445 Received: from murder ([unix socket])
1448 Received: from salemslot.mr.itd.umich.edu (salemslot.mr.itd.umich.edu
[14
1.211.14.58]) 1451 Received: FROM paploo.uhi.ac.uk
(appl.prod.collab.uhi.ac.uk [194.35.219.18
4])
1454 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1460 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
1464 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1467 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1470 Received: (from apache@localhost)
1476 From: stephen.marquard@uct.ac.za
1495 SAK-6216 merge event change from SAK-11169 (r39033) to synchronize
branch
with 2-5-x (for convenience for UCT local build) 1498 This automatic
notification message was sent by Sakai Collab (https://collab.sakaiproject.org/portal) from the Source site.
1503 From louis@media.berkeley.edu Thu Jan 3 19:51:21 2008
1505 Received: from murder (mail.umich.edu [141.211.14.91])
1509 Received: from murder ([unix socket])
1512 Received: from eyewitness.mr.itd.umich.edu (eyewitness.mr.itd.umich.edu
[1
41.211.93.142]) 1515 Received: FROM paploo.uhi.ac.uk
(appl.prod.collab.uhi.ac.uk [194.35.219.18
4])
1518 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1524 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
1528 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1531 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1534 Received: (from apache@localhost)
1540 From: louis@media.berkeley.edu
1563 This automatic notification message was sent by Sakai Collab
(https://collab.sakaiproject.org/portal) from the Source site.
1568 From louis@media.berkeley.edu Thu Jan 3 17:18:23 2008

1570 Received: from murder (mail.umich.edu [141.211.14.91])
1574 Received: from murder ([unix socket])
1577 Received: from salemslot.mr.itd.umich.edu (salemslot.mr.itd.umich.edu
[14
1.211.14.58]) 1580 Received: FROM paploo.uhi.ac.uk
(appl.prod.collab.uhi.ac.uk [194.35.219.18
4])
1583 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1589 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
1593 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1596 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1599 Received: (from apache@localhost)
1605 From: louis@media.berkeley.edu
1628 This automatic notification message was sent by Sakai Collab
(<https://collab.sakaiproject.org/portal>) from the Source site.
1633 From ray@media.berkeley.edu Thu Jan 3 17:07:00 2008
1635 Received: from murder (mail.umich.edu [141.211.14.39])
1639 Received: from murder ([unix socket])
1642 Received: from anniehall.mr.itd.umich.edu (anniehall.mr.itd.umich.edu
[14
1.211.93.141]) 1645 Received: FROM paploo.uhi.ac.uk
(appl.prod.collab.uhi.ac.uk [194.35.219.18
4])
1648 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1654 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
1658 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1661 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1664 Received: (from apache@localhost)
1670 From: ray@media.berkeley.edu
1692 This automatic notification message was sent by Sakai Collab
(<https://collab.sakaiproject.org/portal>) from the Source site. 1697 From
cwen@iupui.edu Thu Jan 3 16:34:40 2008
1699 Received: from murder (mail.umich.edu [141.211.14.34])
1703 Received: from murder ([unix socket])
1706 Received: from icestorm.mr.itd.umich.edu (icestorm.mr.itd.umich.edu
[141.2
11.93.149]) 1709 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
[194.35.219.18
4])
1712 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
1718 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
1722 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
[134.68.22
0.122]) 1725 Received: from nakamura.uits.iupui.edu (localhost
[127.0.0.1])
1728 Received: (from apache@localhost)
1734 From: cwen@iupui.edu

1757 This automatic notification message was sent by Sakai Collab
 (https://collab.sakaiproject.org/portal) from the Source site. 1762 From
 cwen@iupui.edu Thu Jan 3 16:29:07 2008
 1764 Received: from murder (mail.umich.edu [141.211.14.46])
 1768 Received: from murder ([unix socket])
 1771 Received: from galaxyquest.mr.itd.umich.edu
 (galaxyquest.mr.itd.umich.edu
 [141.211.93.145])
 1774 Received: FROM paploo.uhi.ac.uk (appl.prod.collab.uhi.ac.uk
 [194.35.219.18
 4])
 1777 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
 1783 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
 1787 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
 [134.68.22
 0.122]) 1790 Received: from nakamura.uits.iupui.edu (localhost
 [127.0.0.1])
 1793 Received: (from apache@localhost)
 1799 From: cwen@iupui.edu
 1832 This automatic notification message was sent by Sakai Collab
 (https://collab.sakaiproject.org/portal) from the Source site. 1837 From
 cwen@iupui.edu Thu Jan 3 16:23:48 2008
 1839 Received: from murder (mail.umich.edu [141.211.14.91])
 1843 Received: from murder ([unix socket])
 1846 Received: from salemslot.mr.itd.umich.edu (salemslot.mr.itd.umich.edu
 [14
 1.211.14.58]) 1849 Received: FROM paploo.uhi.ac.uk
 (appl.prod.collab.uhi.ac.uk [194.35.219.18
 4])
 1852 Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
 1858 Received: from prod.collab.uhi.ac.uk ([194.35.219.182])
 1862 Received: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu
 [134.68.22
 0.122]) 1865 Received: from nakamura.uits.iupui.edu (localhost
 [127.0.0.1])
 1868 Received: (from apache@localhost)
 1874 From: cwen@iupui.edu
 1907 This automatic notification message was sent by Sakai Collab
 (https://collab.sakaiproject.org/portal) from the Source site.

```
In [101... # Search for lines that start with 'From'
for i, line in enumerate(lines):
    if line.startswith("From"): print(i, line.strip())
    #print line number and line
```

0 From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008
37 From: stephen.marquard@uct.ac.za
65 From louis@media.berkeley.edu Fri Jan 4 18:10:48 2008
102 From: louis@media.berkeley.edu
130 From zqian@umich.edu Fri Jan 4 16:10:39 2008
167 From: zqian@umich.edu
194 From rjlowe@iupui.edu Fri Jan 4 15:46:24 2008
231 From: rjlowe@iupui.edu
262 From zqian@umich.edu Fri Jan 4 15:03:18 2008
299 From: zqian@umich.edu
335 From rjlowe@iupui.edu Fri Jan 4 14:50:18 2008
372 From: rjlowe@iupui.edu
414 From cwen@iupui.edu Fri Jan 4 11:37:30 2008
451 From: cwen@iupui.edu
493 From cwen@iupui.edu Fri Jan 4 11:35:08 2008
530 From: cwen@iupui.edu
572 From gsilver@umich.edu Fri Jan 4 11:12:37 2008
609 From: gsilver@umich.edu
638 From gsilver@umich.edu Fri Jan 4 11:11:52 2008
675 From: gsilver@umich.edu
704 From zqian@umich.edu Fri Jan 4 11:11:03 2008
741 From: zqian@umich.edu
769 From gsilver@umich.edu Fri Jan 4 11:10:22 2008
806 From: gsilver@umich.edu
835 From wagnermr@iupui.edu Fri Jan 4 10:38:42 2008
872 From: wagnermr@iupui.edu
904 From zqian@umich.edu Fri Jan 4 10:17:43 2008
941 From: zqian@umich.edu
969 From antranig@caret.cam.ac.uk Fri Jan 4 10:04:14 2008
1006 From: antranig@caret.cam.ac.uk
1046 From gopal.ramasammycook@gmail.com Fri Jan 4 09:05:31 2008
1083 From: gopal.ramasammycook@gmail.com
1120 From david.horwitz@uct.ac.za Fri Jan 4 07:02:32 2008
1157 From: david.horwitz@uct.ac.za
1206 From david.horwitz@uct.ac.za Fri Jan 4 06:08:27 2008
1243 From: david.horwitz@uct.ac.za
1275 From david.horwitz@uct.ac.za Fri Jan 4 04:49:08 2008
1312 From: david.horwitz@uct.ac.za
1353 From david.horwitz@uct.ac.za Fri Jan 4 04:33:44 2008
1390 From: david.horwitz@uct.ac.za
1439 From stephen.marquard@uct.ac.za Fri Jan 4 04:07:34 2008
1476 From: stephen.marquard@uct.ac.za
1503 From louis@media.berkeley.edu Thu Jan 3 19:51:21 2008
1540 From: louis@media.berkeley.edu
1568 From louis@media.berkeley.edu Thu Jan 3 17:18:23 2008
1605 From: louis@media.berkeley.edu
1633 From ray@media.berkeley.edu Thu Jan 3 17:07:00 2008
1670 From: ray@media.berkeley.edu
1697 From cwen@iupui.edu Thu Jan 3 16:34:40 2008
1734 From: cwen@iupui.edu
1762 From cwen@iupui.edu Thu Jan 3 16:29:07 2008
1799 From: cwen@iupui.edu
1837 From cwen@iupui.edu Thu Jan 3 16:23:48 2008

```
# Search for lines that start with 'F', followed by 2 characters, followed by
import re

pattern = r"^F..m:"
for(i, line) in enumerate(lines, start=1):
    if re.match(pattern, line):
        print(i, line.strip())
```

1874 From: cwen@iupui.edu

In [102...

```
38 From: stephen.marquard@uct.ac.za
103 From: louis@media.berkeley.edu
168 From: zqian@umich.edu
232 From: rjlowe@iupui.edu
300 From: zqian@umich.edu
373 From: rjlowe@iupui.edu
452 From: cwen@iupui.edu
531 From: cwen@iupui.edu
610 From: gsilver@umich.edu
676 From: gsilver@umich.edu
742 From: zqian@umich.edu
807 From: gsilver@umich.edu
873 From: wagnermr@iupui.edu
942 From: zqian@umich.edu
1007 From: antranig@caret.cam.ac.uk
1084 From: gopal.ramasammycook@gmail.com
1158 From: david.horwitz@uct.ac.za
1244 From: david.horwitz@uct.ac.za
1313 From: david.horwitz@uct.ac.za
1391 From: david.horwitz@uct.ac.za
1477 From: stephen.marquard@uct.ac.za
1541 From: louis@media.berkeley.edu
1606 From: louis@media.berkeley.edu
1671 From: ray@media.berkeley.edu
1735 From: cwen@iupui.edu
1800 From: cwen@iupui.edu
1875 From: cwen@iupui.edu
```

In [103...

```
# Search for lines that start with From and have an at sign
pattern = r"^From.*@"

for i, line in enumerate(lines, start=1):
    if re.match(pattern, line):
        print(i, line.strip())
```

1 From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008

38 From: stephen.marquard@uct.ac.za
66 From louis@media.berkeley.edu Fri Jan 4 18:10:48 2008
103 From: louis@media.berkeley.edu
131 From zqian@umich.edu Fri Jan 4 16:10:39 2008
168 From: zqian@umich.edu
195 From rjlowe@iupui.edu Fri Jan 4 15:46:24 2008
232 From: rjlowe@iupui.edu
263 From zqian@umich.edu Fri Jan 4 15:03:18 2008
300 From: zqian@umich.edu
336 From rjlowe@iupui.edu Fri Jan 4 14:50:18 2008
373 From: rjlowe@iupui.edu
415 From cwen@iupui.edu Fri Jan 4 11:37:30 2008
452 From: cwen@iupui.edu
494 From cwen@iupui.edu Fri Jan 4 11:35:08 2008
531 From: cwen@iupui.edu
573 From gsilver@umich.edu Fri Jan 4 11:12:37 2008
610 From: gsilver@umich.edu
639 From gsilver@umich.edu Fri Jan 4 11:11:52 2008
676 From: gsilver@umich.edu
705 From zqian@umich.edu Fri Jan 4 11:11:03 2008
742 From: zqian@umich.edu
770 From gsilver@umich.edu Fri Jan 4 11:10:22 2008
807 From: gsilver@umich.edu
836 From wagnermr@iupui.edu Fri Jan 4 10:38:42 2008
873 From: wagnermr@iupui.edu
905 From zqian@umich.edu Fri Jan 4 10:17:43 2008
942 From: zqian@umich.edu
970 From antranig@caret.cam.ac.uk Fri Jan 4 10:04:14 2008
1007 From: antranig@caret.cam.ac.uk
1047 From gopal.ramasammycook@gmail.com Fri Jan 4 09:05:31 2008
1084 From: gopal.ramasammycook@gmail.com
1121 From david.horwitz@uct.ac.za Fri Jan 4 07:02:32 2008
1158 From: david.horwitz@uct.ac.za
1207 From david.horwitz@uct.ac.za Fri Jan 4 06:08:27 2008
1244 From: david.horwitz@uct.ac.za
1276 From david.horwitz@uct.ac.za Fri Jan 4 04:49:08 2008
1313 From: david.horwitz@uct.ac.za
1354 From david.horwitz@uct.ac.za Fri Jan 4 04:33:44 2008
1391 From: david.horwitz@uct.ac.za
1440 From stephen.marquard@uct.ac.za Fri Jan 4 04:07:34 2008
1477 From: stephen.marquard@uct.ac.za
1504 From louis@media.berkeley.edu Thu Jan 3 19:51:21 2008
1541 From: louis@media.berkeley.edu
1569 From louis@media.berkeley.edu Thu Jan 3 17:18:23 2008
1606 From: louis@media.berkeley.edu
1634 From ray@media.berkeley.edu Thu Jan 3 17:07:00 2008
1671 From: ray@media.berkeley.edu
1698 From cwen@iupui.edu Thu Jan 3 16:34:40 2008
1735 From: cwen@iupui.edu
1763 From cwen@iupui.edu Thu Jan 3 16:29:07 2008
1800 From: cwen@iupui.edu
1838 From cwen@iupui.edu Thu Jan 3 16:23:48 2008
1875 From: cwen@iupui.edu

```
In [104... #example of findall method
text = """
From alice@example.com Sat Jan  5 09:14:16 2008
From bob123@mail.org Fri Feb  1 12:22:33 2009
This line has no email
Contact us at support@test.co.in
"""

#pattern for email
pattern = r"[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}"

emails = re.findall(pattern, text)
print(emails)

['alice@example.com', 'bob123@mail.org', 'support@test.co.in']
```

```
In [105... # Search for lines that have an at sign between characters
pattern = r".+@.+"

for i, line in enumerate(lines, start=1):
    if re.search(pattern, line):
        print(i, line.strip())
```

1 From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008
2 Return-Path: <postmaster@collab.sakaiproject.org>
19 Message-ID: <200801051412.m05ECIaH010327@nakamura.uits.iupui.edu>
24 for <source@collab.sakaiproject.org>;
28 for <source@collab.sakaiproject.org>; Sat, 5 Jan 2008 14:13:33 +0000
(GMT)
31 for <source@collab.sakaiproject.org>; Sat, 5 Jan 2008 09:12:19 -0500
32 Received: (from apache@localhost)
34 for source@collab.sakaiproject.org; Sat, 5 Jan 2008 09:12:18 -0500
36 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
step
hen.marquard@uct.ac.za using -f
37 To: source@collab.sakaiproject.org
38 From: stephen.marquard@uct.ac.za
50 Author: stephen.marquard@uct.ac.za
66 From louis@media.berkeley.edu Fri Jan 4 18:10:48 2008
67 Return-Path: <postmaster@collab.sakaiproject.org>
84 Message-ID: <200801042308.m04N8v60008125@nakamura.uits.iupui.edu>
89 for <source@collab.sakaiproject.org>;
93 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 23:10:10 +0000
(GMT)
96 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 18:08:57 -0500
97 Received: (from apache@localhost)
99 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 18:08:57 -0500
101 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
lou
is@media.berkeley.edu using -f
102 To: source@collab.sakaiproject.org
103 From: louis@media.berkeley.edu
115 Author: louis@media.berkeley.edu
131 From zqian@umich.edu Fri Jan 4 16:10:39 2008
132 Return-Path: <postmaster@collab.sakaiproject.org>
149 Message-ID: <200801042109.m04L92hb007923@nakamura.uits.iupui.edu>
154 for <source@collab.sakaiproject.org>;
158 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 21:10:14 +0000
(GMT)
161 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:09:02 -0500
162 Received: (from apache@localhost)
164 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 16:09:02 -0500
166 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
zqi
an@umich.edu using -f
167 To: source@collab.sakaiproject.org
168 From: zqian@umich.edu
180 Author: zqian@umich.edu
195 From rjlowe@iupui.edu Fri Jan 4 15:46:24 2008
196 Return-Path: <postmaster@collab.sakaiproject.org>
213 Message-ID: <200801042044.m04Kiem3007881@nakamura.uits.iupui.edu>
218 for <source@collab.sakaiproject.org>;
222 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 20:45:52 +0000
(GMT)
225 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:44:40 -0500
226 Received: (from apache@localhost)

228 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 15:44:40 -0500
230 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
rjl
owe@iupui.edu using -f
231 To: source@collab.sakaiproject.org
232 From: rjlowe@iupui.edu
244 Author: rjlowe@iupui.edu
263 From zqian@umich.edu Fri Jan 4 15:03:18 2008
264 Return-Path: <postmaster@collab.sakaiproject.org>
281 Message-ID: <200801042001.m04KlcO0007738@nakamura.uits.iupui.edu>
286 for <source@collab.sakaiproject.org>;
290 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 20:02:50 +0000
(GMT)
293 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:01:38 -0500
294 Received: (from apache@localhost)
296 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 15:01:38 -0500
298 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
zqi
an@umich.edu using -f
299 To: source@collab.sakaiproject.org
300 From: zqian@umich.edu
312 Author: zqian@umich.edu
321 Sakai Source Repository #38024 Wed Nov 07 14:54:46 MST
20 07 zqian@umich.edu Fix to SAK-10788: If a provided id in
a course site is fake or doesn't provide any user information, Site Info appears
to
be like project site with empty participant list
336 From rjlowe@iupui.edu Fri Jan 4 14:50:18
2008
337 Return-Path: <postmaster@collab.sakaiproject.org>
354 Message-ID: <200801041948.m04JmdwO007705@nakamura.uits.iupui.edu>
359 for <source@collab.sakaiproject.org>;
363 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 19:49:51 +0000
(GMT)
366 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 14:48:40 -0500
367 Received: (from apache@localhost)
369 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 14:48:39 -0500
371 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
rjl
owe@iupui.edu using -f
372 To: source@collab.sakaiproject.org
373 From: rjlowe@iupui.edu
385 Author: rjlowe@iupui.edu
415 From cwen@iupui.edu Fri Jan 4 11:37:30 2008
416 Return-Path: <postmaster@collab.sakaiproject.org>
433 Message-ID: <200801041635.m04GZQGZ007313@nakamura.uits.iupui.edu>
438 for <source@collab.sakaiproject.org>;
442 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:36:37 +0000
(GMT)
445 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:35:26 -0500
446 Received: (from apache@localhost)
448 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:35:26 -0500

450 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
cwe
n@iupui.edu using -f
451 To: source@collab.sakaiproject.org
452 From: cwen@iupui.edu
464 Author: cwen@iupui.edu
480 r39558 | hu2@iupui.edu | 2007-12-20 15:25:38 -0500 (Thu, 20 Dec 2007) | 3
1
ines
494 From cwen@iupui.edu Fri Jan 4 11:35:08 2008
495 Return-Path: <postmaster@collab.sakaiproject.org>
512 Message-ID: <200801041633.m04GX6eG007292@nakamura.uits.iupui.edu>
517 for <source@collab.sakaiproject.org>;
521 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:34:17 +0000
(GMT)
524 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:33:06 -0500
525 Received: (from apache@localhost)
527 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:33:06 -0500
529 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
cwe
n@iupui.edu using -f
530 To: source@collab.sakaiproject.org
531 From: cwen@iupui.edu
543 Author: cwen@iupui.edu
559 r39571 | hu2@iupui.edu | 2007-12-20 21:26:28 -0500 (Thu, 20 Dec 2007) | 3
1
ines
573 From gsilver@umich.edu Fri Jan 4 11:12:37 2008
574 Return-Path: <postmaster@collab.sakaiproject.org>
591 Message-ID: <200801041611.m04GB1Lb007221@nakamura.uits.iupui.edu>
596 for <source@collab.sakaiproject.org>;
600 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:12:12 +0000
(GMT)
603 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:11:01 -0500
604 Received: (from apache@localhost)
606 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:11:01 -0500
608 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
gsi
lver@umich.edu using -f
609 To: source@collab.sakaiproject.org
610 From: gsilver@umich.edu
622 Author: gsilver@umich.edu
639 From gsilver@umich.edu Fri Jan 4 11:11:52 2008
640 Return-Path: <postmaster@collab.sakaiproject.org>
657 Message-ID: <200801041610.m04GA5KP007209@nakamura.uits.iupui.edu>
662 for <source@collab.sakaiproject.org>;
666 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:11:16 +0000
(GMT)
669 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:10:05 -0500
670 Received: (from apache@localhost)
672 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:10:05 -0500
674 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
gsi

lver@umich.edu using -f
675 To: source@collab.sakaiproject.org
676 From: gsilver@umich.edu
688 Author: gsilver@umich.edu
705 From zqian@umich.edu Fri Jan 4 11:11:03 2008
706 Return-Path: <postmaster@collab.sakaiproject.org>
723 Message-ID: <200801041609.m04G9EuX007197@nakamura.uits.iupui.edu>
728 for <source@collab.sakaiproject.org>;
732 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:10:26 +0000
(GMT)
735 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:09:15 -0500
736 Received: (from apache@localhost)
738 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:09:14 -0500
740 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
zqi
an@umich.edu using -f
741 To: source@collab.sakaiproject.org
742 From: zqian@umich.edu
754 Author: zqian@umich.edu
770 From gsilver@umich.edu Fri Jan 4 11:10:22 2008
771 Return-Path: <postmaster@collab.sakaiproject.org>
788 Message-ID: <200801041608.m04G8d7w007184@nakamura.uits.iupui.edu>
793 for <source@collab.sakaiproject.org>;
797 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:09:50 +0000
(GMT)
800 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:08:39 -0500
801 Received: (from apache@localhost)
803 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:08:39 -0500
805 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
gsi
lver@umich.edu using -f
806 To: source@collab.sakaiproject.org
807 From: gsilver@umich.edu
819 Author: gsilver@umich.edu
836 From wagnermr@iupui.edu Fri Jan 4 10:38:42 2008
837 Return-Path: <postmaster@collab.sakaiproject.org>
854 Message-ID: <200801041537.m04Fb6Ci007092@nakamura.uits.iupui.edu>
859 for <source@collab.sakaiproject.org>;
863 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:38:17 +0000
(GMT)
866 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 10:37:06 -0500
867 Received: (from apache@localhost)
869 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 10:37:06 -0500
871 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
wag
nermr@iupui.edu using -f
872 To: source@collab.sakaiproject.org
873 From: wagnermr@iupui.edu
885 Author: wagnermr@iupui.edu
905 From zqian@umich.edu Fri Jan 4 10:17:43 2008
906 Return-Path: <postmaster@collab.sakaiproject.org>
923 Message-ID: <200801041515.m04FFv42007050@nakamura.uits.iupui.edu>
928 for <source@collab.sakaiproject.org>;

932 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:17:08 +0000
(GMT)
935 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 10:15:57 -0500
936 Received: (from apache@localhost)
938 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 10:15:57 -0500
940 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
zqi
an@umich.edu using -f
941 To: source@collab.sakaiproject.org
942 From: zqian@umich.edu
954 Author: zqian@umich.edu
970 From antranig@caret.cam.ac.uk Fri Jan 4 10:04:14 2008
971 Return-Path: <postmaster@collab.sakaiproject.org>
988 Message-ID: <200801041502.m04F21Jo007031@nakamura.uits.iupui.edu>
993 for <source@collab.sakaiproject.org>;
997 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:03:12 +0000
(GMT)
1000 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 10:02:01 -0500
1001 Received: (from apache@localhost)
1003 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 10:02:01 -0500
1005 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
an
tranig@caret.cam.ac.uk using -f
1006 To: source@collab.sakaiproject.org
1007 From: antranig@caret.cam.ac.uk
1019 Author: antranig@caret.cam.ac.uk
1047 From gopal.ramasammycook@gmail.com Fri Jan 4 09:05:31 2008
1048 Return-Path: <postmaster@collab.sakaiproject.org>
1065 Message-ID: <200801041403.m04E3psW006926@nakamura.uits.iupui.edu>
1070 for <source@collab.sakaiproject.org>;
1074 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 14:05:03 +0000
(GM
T) 1077 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 09:03:52
-0500
1078 Received: (from apache@localhost)
1080 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 09:03:51 -0500
1082 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
go
pal.ramasammycook@gmail.com using -f
1083 To: source@collab.sakaiproject.org
1084 From: gopal.ramasammycook@gmail.com
1096 Author: gopal.ramasammycook@gmail.com
1121 From david.horwitz@uct.ac.za Fri Jan 4 07:02:32 2008
1122 Return-Path: <postmaster@collab.sakaiproject.org>
1139 Message-ID: <200801041200.m04C0gfK006793@nakamura.uits.iupui.edu>
1144 for <source@collab.sakaiproject.org>;
1148 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 12:01:53 +0000
(GM
T) 1151 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 07:00:42
-0500
1152 Received: (from apache@localhost)
1154 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 07:00:42 -0500

1156 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
da
vid.horwitz@uct.ac.za using -f
1157 To: source@collab.sakaiproject.org
1158 From: david.horwitz@uct.ac.za
1170 Author: david.horwitz@uct.ac.za
1186 r39753 | david.horwitz@uct.ac.za | 2008-01-04 13:05:51 +0200 (Fri, 04
Jan
2008) | 1 line
1190 dhorwitz@david-horwitz-6:~/branchManagemnt/sakai_2-5-x> svn merge -
c39753
<https://source.sakaiproject.org/svn/polls/trunk/polls/>
1197 dhorwitz@david-horwitz-6:~/branchManagemnt/sakai_2-5-x> svn resolved
poll
s/tool/src/webapp/WEB-INF/requestContext.xml
1207 From david.horwitz@uct.ac.za Fri Jan 4 06:08:27 2008
1208 Return-Path: <postmaster@collab.sakaiproject.org>
1225 Message-ID: <200801041106.m04B6lK3006677@nakamura.uits.iupui.edu>
1230 for <source@collab.sakaiproject.org>;
1234 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:07:58 +0000
(GM
T) 1237 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 06:06:47
-0500
1238 Received: (from apache@localhost)
1240 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 06:06:47 -0500
1242 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
da
vid.horwitz@uct.ac.za using -f
1243 To: source@collab.sakaiproject.org
1244 From: david.horwitz@uct.ac.za
1256 Author: david.horwitz@uct.ac.za
1276 From david.horwitz@uct.ac.za Fri Jan 4 04:49:08 2008
1277 Return-Path: <postmaster@collab.sakaiproject.org>
1294 Message-ID: <200801040947.m049lUxo006517@nakamura.uits.iupui.edu>
1299 for <source@collab.sakaiproject.org>;
1303 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 09:48:40 +0000
(GM
T) 1306 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 04:47:30
-0500
1307 Received: (from apache@localhost)
1309 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 04:47:30 -0500
1311 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
da
vid.horwitz@uct.ac.za using -f
1312 To: source@collab.sakaiproject.org
1313 From: david.horwitz@uct.ac.za
1325 Author: david.horwitz@uct.ac.za
1335 r39641 | josrodri@iupui.edu | 2007-12-28 23:44:24 +0200 (Fri, 28 Dec
2007)
| 1 line
1340 dhorwitz@david-horwitz-6:~/branchManagemnt/sakai_2-5-x> svn merge -
c39641

https://source.sakaiproject.org/svn/podcasts/trunk podcasts/
1354 From david.horwitz@uct.ac.za Fri Jan 4 04:33:44 2008
1355 Return-Path: <postmaster@collab.sakaiproject.org>
1372 Message-ID: <200801040932.m049W2i5006493@nakamura.uits.iupui.edu>
1377 for <source@collab.sakaiproject.org>;
1381 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 09:33:13 +0000
(GM
T) 1384 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 04:32:03
-0500
1385 Received: (from apache@localhost)
1387 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 04:32:02 -0500
1389 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
da
vid.horwitz@uct.ac.za using -f
1390 To: source@collab.sakaiproject.org
1391 From: david.horwitz@uct.ac.za
1403 Author: david.horwitz@uct.ac.za
1419 r39146 | josrodri@iupui.edu | 2007-12-12 21:40:33 +0200 (Wed, 12 Dec
2007)
| 1 line
1423 dhorwitz@david-horwitz-6:~/branchManagemnt/sakai_2-5-x> svn merge -
c39146
https://source.sakaiproject.org/svn/podcasts/trunk podcasts/
1440 From stephen.marquard@uct.ac.za Fri Jan 4 04:07:34 2008
1441 Return-Path: <postmaster@collab.sakaiproject.org>
1458 Message-ID: <200801040905.m0495rWB006420@nakamura.uits.iupui.edu>
1463 for <source@collab.sakaiproject.org>;
1467 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 09:07:04 +0000
(GM
T) 1470 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 04:05:54
-0500
1471 Received: (from apache@localhost)
1473 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 04:05:53 -0500
1475 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
st
ephen.marquard@uct.ac.za using -f
1476 To: source@collab.sakaiproject.org
1477 From: stephen.marquard@uct.ac.za
1489 Author: stephen.marquard@uct.ac.za
1504 From louis@media.berkeley.edu Thu Jan 3 19:51:21 2008
1505 Return-Path: <postmaster@collab.sakaiproject.org>
1522 Message-ID: <200801040023.m040NpCc005473@nakamura.uits.iupui.edu>
1527 for <source@collab.sakaiproject.org>;
1531 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 00:25:00 +0000
(GM
T) 1534 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 19:23:51
-0500
1535 Received: (from apache@localhost)
1537 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 19:23:51 -0500
1539 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
lo
uis@media.berkeley.edu using -f
1540 To: source@collab.sakaiproject.org

1541 From: louis@media.berkeley.edu
1553 Author: louis@media.berkeley.edu
1569 From louis@media.berkeley.edu Thu Jan 3 17:18:23 2008
1570 Return-Path: <postmaster@collab.sakaiproject.org>
1587 Message-ID: <200801032216.m03MGhDa005292@nakamura.uits.iupui.edu>
1592 for <source@collab.sakaiproject.org>;
1596 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 22:17:52 +0000
(GM
T) 1599 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 17:16:43
-0500
1600 Received: (from apache@localhost)
1602 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 17:16:43 -0500
1604 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
lo
uis@media.berkeley.edu using -f
1605 To: source@collab.sakaiproject.org
1606 From: louis@media.berkeley.edu
1618 Author: louis@media.berkeley.edu
1634 From ray@media.berkeley.edu Thu Jan 3 17:07:00 2008
1635 Return-Path: <postmaster@collab.sakaiproject.org>
1652 Message-ID: <200801032205.m03M5Ea7005273@nakamura.uits.iupui.edu>
1657 for <source@collab.sakaiproject.org>;
1661 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 22:06:23 +0000
(GM
T) 1664 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 17:05:14
-0500
1665 Received: (from apache@localhost)
1667 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 17:05:14 -0500
1669 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
ra
y@media.berkeley.edu using -f
1670 To: source@collab.sakaiproject.org
1671 From: ray@media.berkeley.edu
1683 Author: ray@media.berkeley.edu
1698 From cwen@iupui.edu Thu Jan 3 16:34:40 2008
1699 Return-Path: <postmaster@collab.sakaiproject.org>
1716 Message-ID: <200801032133.m03LX3gG005191@nakamura.uits.iupui.edu>
1721 for <source@collab.sakaiproject.org>;
1725 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 21:34:12 +0000
(GM
T) 1728 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 16:33:03
-0500
1729 Received: (from apache@localhost)
1731 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 16:33:03 -0500
1733 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
cw
en@iupui.edu using -f
1734 To: source@collab.sakaiproject.org
1735 From: cwen@iupui.edu
1747 Author: cwen@iupui.edu
1763 From cwen@iupui.edu Thu Jan 3 16:29:07 2008
1764 Return-Path: <postmaster@collab.sakaiproject.org>
1781 Message-ID: <200801032127.m03LRUqH005177@nakamura.uits.iupui.edu>

```

1786 for <source@collab.sakaiproject.org>;
1790 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 21:28:38 +0000
(GM
T) 1793 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 16:27:30
-0500
1794 Received: (from apache@localhost)
1796 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 16:27:30 -0500
1798 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
    cw
en@iupui.edu using -f
1799 To: source@collab.sakaiproject.org
1800 From: cwen@iupui.edu
1812 Author: cwen@iupui.edu
1824 r39403 | wagnermr@iupui.edu | 2007-12-17 17:11:08 -0500 (Mon, 17 Dec
2007)
| 3 lines
1838 From cwen@iupui.edu Thu Jan 3 16:23:48 2008
1839 Return-Path: <postmaster@collab.sakaiproject.org>
1856 Message-ID: <200801032122.m03LMFo4005148@nakamura.uits.iupui.edu>
1861 for <source@collab.sakaiproject.org>;
1865 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 21:23:24 +0000
(GM
T) 1868 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 16:22:15
-0500
1869 Received: (from apache@localhost)
1871 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 16:22:15 -0500
1873 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
    cw
en@iupui.edu using -f
1874 To: source@collab.sakaiproject.org
1875 From: cwen@iupui.edu
1887 Author: cwen@iupui.edu
1899 r35014 | wagnermr@iupui.edu | 2007-09-12 16:17:59 -0400 (Wed, 12 Sep
2007)
| 3 lines

```

```

In [106... # Search for lines that have an at sign between characters
           # The characters must be a letter or number
pattern = r"[A-Za-z0-9]@[A-Za-z0-9]"

for i, line in enumerate(lines, start=1):
    if re.search(pattern, line):
        print(i, line.strip())

```

1 From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008
2 Return-Path: <postmaster@collab.sakaiproject.org>
19 Message-ID: <200801051412.m05ECIaH010327@nakamura.uits.iupui.edu>
24 for <source@collab.sakaiproject.org>;
28 for <source@collab.sakaiproject.org>; Sat, 5 Jan 2008 14:13:33 +0000
(GMT)
31 for <source@collab.sakaiproject.org>; Sat, 5 Jan 2008 09:12:19 -0500
32 Received: (from apache@localhost)
34 for source@collab.sakaiproject.org; Sat, 5 Jan 2008 09:12:18 -0500
36 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
step
hen.marquard@uct.ac.za using -f
37 To: source@collab.sakaiproject.org
38 From: stephen.marquard@uct.ac.za
50 Author: stephen.marquard@uct.ac.za
66 From louis@media.berkeley.edu Fri Jan 4 18:10:48 2008
67 Return-Path: <postmaster@collab.sakaiproject.org>
84 Message-ID: <200801042308.m04N8v60008125@nakamura.uits.iupui.edu>
89 for <source@collab.sakaiproject.org>;
93 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 23:10:10 +0000
(GMT)
96 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 18:08:57 -0500
97 Received: (from apache@localhost)
99 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 18:08:57 -0500
101 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
lou
is@media.berkeley.edu using -f
102 To: source@collab.sakaiproject.org
103 From: louis@media.berkeley.edu
115 Author: louis@media.berkeley.edu
131 From zqian@umich.edu Fri Jan 4 16:10:39 2008
132 Return-Path: <postmaster@collab.sakaiproject.org>
149 Message-ID: <200801042109.m04L92hb007923@nakamura.uits.iupui.edu>
154 for <source@collab.sakaiproject.org>;
158 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 21:10:14 +0000
(GMT)
161 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:09:02 -0500
162 Received: (from apache@localhost)
164 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 16:09:02 -0500
166 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
zqi
an@umich.edu using -f
167 To: source@collab.sakaiproject.org
168 From: zqian@umich.edu
180 Author: zqian@umich.edu
195 From rjlowe@iupui.edu Fri Jan 4 15:46:24 2008
196 Return-Path: <postmaster@collab.sakaiproject.org>
213 Message-ID: <200801042044.m04Kiem3007881@nakamura.uits.iupui.edu>
218 for <source@collab.sakaiproject.org>;
222 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 20:45:52 +0000
(GMT)
225 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:44:40 -0500
226 Received: (from apache@localhost)

228 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 15:44:40 -0500
230 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
rjl
owe@iupui.edu using -f
231 To: source@collab.sakaiproject.org
232 From: rjlowe@iupui.edu
244 Author: rjlowe@iupui.edu
263 From zqian@umich.edu Fri Jan 4 15:03:18 2008
264 Return-Path: <postmaster@collab.sakaiproject.org>
281 Message-ID: <200801042001.m04KlcO0007738@nakamura.uits.iupui.edu>
286 for <source@collab.sakaiproject.org>;
290 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 20:02:50 +0000
(GMT)
293 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:01:38 -0500
294 Received: (from apache@localhost)
296 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 15:01:38 -0500
298 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
zqi
an@umich.edu using -f
299 To: source@collab.sakaiproject.org
300 From: zqian@umich.edu
312 Author: zqian@umich.edu
321 Sakai Source Repository #38024 Wed Nov 07 14:54:46 MST
20 07 zqian@umich.edu Fix to SAK-10788: If a provided id in
a course site is fake or doesn't provide any user information, Site Info appears
to
be like project site with empty participant list
336 From rjlowe@iupui.edu Fri Jan 4 14:50:18
2008
337 Return-Path: <postmaster@collab.sakaiproject.org>
354 Message-ID: <200801041948.m04JmdwO007705@nakamura.uits.iupui.edu>
359 for <source@collab.sakaiproject.org>;
363 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 19:49:51 +0000
(GMT)
366 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 14:48:40 -0500
367 Received: (from apache@localhost)
369 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 14:48:39 -0500
371 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
rjl
owe@iupui.edu using -f
372 To: source@collab.sakaiproject.org
373 From: rjlowe@iupui.edu
385 Author: rjlowe@iupui.edu
415 From cwen@iupui.edu Fri Jan 4 11:37:30 2008
416 Return-Path: <postmaster@collab.sakaiproject.org>
433 Message-ID: <200801041635.m04GZQGZ007313@nakamura.uits.iupui.edu>
438 for <source@collab.sakaiproject.org>;
442 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:36:37 +0000
(GMT)
445 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:35:26 -0500
446 Received: (from apache@localhost)
448 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:35:26 -0500

450 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
cwe
n@iupui.edu using -f
451 To: source@collab.sakaiproject.org
452 From: cwen@iupui.edu
464 Author: cwen@iupui.edu
480 r39558 | hu2@iupui.edu | 2007-12-20 15:25:38 -0500 (Thu, 20 Dec 2007) | 3
1
ines
494 From cwen@iupui.edu Fri Jan 4 11:35:08 2008
495 Return-Path: <postmaster@collab.sakaiproject.org>
512 Message-ID: <200801041633.m04GX6eG007292@nakamura.uits.iupui.edu>
517 for <source@collab.sakaiproject.org>;
521 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:34:17 +0000
(GMT)
524 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:33:06 -0500
525 Received: (from apache@localhost)
527 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:33:06 -0500
529 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
cwe
n@iupui.edu using -f
530 To: source@collab.sakaiproject.org
531 From: cwen@iupui.edu
543 Author: cwen@iupui.edu
559 r39571 | hu2@iupui.edu | 2007-12-20 21:26:28 -0500 (Thu, 20 Dec 2007) | 3
1
ines
573 From gsilver@umich.edu Fri Jan 4 11:12:37 2008
574 Return-Path: <postmaster@collab.sakaiproject.org>
591 Message-ID: <200801041611.m04GB1Lb007221@nakamura.uits.iupui.edu>
596 for <source@collab.sakaiproject.org>;
600 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:12:12 +0000
(GMT)
603 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:11:01 -0500
604 Received: (from apache@localhost)
606 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:11:01 -0500
608 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
gsi
lver@umich.edu using -f
609 To: source@collab.sakaiproject.org
610 From: gsilver@umich.edu
622 Author: gsilver@umich.edu
639 From gsilver@umich.edu Fri Jan 4 11:11:52 2008
640 Return-Path: <postmaster@collab.sakaiproject.org>
657 Message-ID: <200801041610.m04GA5KP007209@nakamura.uits.iupui.edu>
662 for <source@collab.sakaiproject.org>;
666 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:11:16 +0000
(GMT)
669 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:10:05 -0500
670 Received: (from apache@localhost)
672 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:10:05 -0500
674 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
gsi

lver@umich.edu using -f
675 To: source@collab.sakaiproject.org
676 From: gsilver@umich.edu
688 Author: gsilver@umich.edu
705 From zqian@umich.edu Fri Jan 4 11:11:03 2008
706 Return-Path: <postmaster@collab.sakaiproject.org>
723 Message-ID: <200801041609.m04G9EuX007197@nakamura.uits.iupui.edu>
728 for <source@collab.sakaiproject.org>;
732 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:10:26 +0000
(GMT)
735 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:09:15 -0500
736 Received: (from apache@localhost)
738 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:09:14 -0500
740 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
zqi
an@umich.edu using -f
741 To: source@collab.sakaiproject.org
742 From: zqian@umich.edu
754 Author: zqian@umich.edu
770 From gsilver@umich.edu Fri Jan 4 11:10:22 2008
771 Return-Path: <postmaster@collab.sakaiproject.org>
788 Message-ID: <200801041608.m04G8d7w007184@nakamura.uits.iupui.edu>
793 for <source@collab.sakaiproject.org>;
797 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 16:09:50 +0000
(GMT)
800 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:08:39 -0500
801 Received: (from apache@localhost)
803 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 11:08:39 -0500
805 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
gsi
lver@umich.edu using -f
806 To: source@collab.sakaiproject.org
807 From: gsilver@umich.edu
819 Author: gsilver@umich.edu
836 From wagnermr@iupui.edu Fri Jan 4 10:38:42 2008
837 Return-Path: <postmaster@collab.sakaiproject.org>
854 Message-ID: <200801041537.m04Fb6Ci007092@nakamura.uits.iupui.edu>
859 for <source@collab.sakaiproject.org>;
863 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:38:17 +0000
(GMT)
866 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 10:37:06 -0500
867 Received: (from apache@localhost)
869 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 10:37:06 -0500
871 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
wag
nermr@iupui.edu using -f
872 To: source@collab.sakaiproject.org
873 From: wagnermr@iupui.edu
885 Author: wagnermr@iupui.edu
905 From zqian@umich.edu Fri Jan 4 10:17:43 2008
906 Return-Path: <postmaster@collab.sakaiproject.org>
923 Message-ID: <200801041515.m04FFv42007050@nakamura.uits.iupui.edu>
928 for <source@collab.sakaiproject.org>;

932 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:17:08 +0000
(GMT)
935 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 10:15:57 -0500
936 Received: (from apache@localhost)
938 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 10:15:57 -0500
940 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
zqi
an@umich.edu using -f
941 To: source@collab.sakaiproject.org
942 From: zqian@umich.edu
954 Author: zqian@umich.edu
970 From antranig@caret.cam.ac.uk Fri Jan 4 10:04:14 2008
971 Return-Path: <postmaster@collab.sakaiproject.org>
988 Message-ID: <200801041502.m04F21Jo007031@nakamura.uits.iupui.edu>
993 for <source@collab.sakaiproject.org>;
997 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 15:03:12 +0000
(GMT)
1000 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 10:02:01 -0500
1001 Received: (from apache@localhost)
1003 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 10:02:01 -0500
1005 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
an
tranig@caret.cam.ac.uk using -f
1006 To: source@collab.sakaiproject.org
1007 From: antranig@caret.cam.ac.uk
1019 Author: antranig@caret.cam.ac.uk
1047 From gopal.ramasammycook@gmail.com Fri Jan 4 09:05:31 2008
1048 Return-Path: <postmaster@collab.sakaiproject.org>
1065 Message-ID: <200801041403.m04E3psW006926@nakamura.uits.iupui.edu>
1070 for <source@collab.sakaiproject.org>;
1074 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 14:05:03 +0000
(GM
T) 1077 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 09:03:52
-0500
1078 Received: (from apache@localhost)
1080 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 09:03:51 -0500
1082 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
go
pal.ramasammycook@gmail.com using -f
1083 To: source@collab.sakaiproject.org
1084 From: gopal.ramasammycook@gmail.com
1096 Author: gopal.ramasammycook@gmail.com
1121 From david.horwitz@uct.ac.za Fri Jan 4 07:02:32 2008
1122 Return-Path: <postmaster@collab.sakaiproject.org>
1139 Message-ID: <200801041200.m04C0gfK006793@nakamura.uits.iupui.edu>
1144 for <source@collab.sakaiproject.org>;
1148 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 12:01:53 +0000
(GM
T) 1151 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 07:00:42
-0500
1152 Received: (from apache@localhost)
1154 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 07:00:42 -0500

1156 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
da
vid.horwitz@uct.ac.za using -f
1157 To: source@collab.sakaiproject.org
1158 From: david.horwitz@uct.ac.za
1170 Author: david.horwitz@uct.ac.za
1186 r39753 | david.horwitz@uct.ac.za | 2008-01-04 13:05:51 +0200 (Fri, 04
Jan
2008) | 1 line
1190 dhorwitz@david-horwitz-6:~/branchManagemnt/sakai_2-5-x> svn merge -
c39753
<https://source.sakaiproject.org/svn/polls/trunk/polls/>
1197 dhorwitz@david-horwitz-6:~/branchManagemnt/sakai_2-5-x> svn resolved
poll
s/tool/src/webapp/WEB-INF/requestContext.xml
1207 From david.horwitz@uct.ac.za Fri Jan 4 06:08:27 2008
1208 Return-Path: <postmaster@collab.sakaiproject.org>
1225 Message-ID: <200801041106.m04B6lK3006677@nakamura.uits.iupui.edu>
1230 for <source@collab.sakaiproject.org>;
1234 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 11:07:58 +0000
(GM
T) 1237 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 06:06:47
-0500
1238 Received: (from apache@localhost)
1240 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 06:06:47 -0500
1242 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
da
vid.horwitz@uct.ac.za using -f
1243 To: source@collab.sakaiproject.org
1244 From: david.horwitz@uct.ac.za
1256 Author: david.horwitz@uct.ac.za
1276 From david.horwitz@uct.ac.za Fri Jan 4 04:49:08 2008
1277 Return-Path: <postmaster@collab.sakaiproject.org>
1294 Message-ID: <200801040947.m049lUxo006517@nakamura.uits.iupui.edu>
1299 for <source@collab.sakaiproject.org>;
1303 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 09:48:40 +0000
(GM
T) 1306 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 04:47:30
-0500
1307 Received: (from apache@localhost)
1309 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 04:47:30 -0500
1311 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
da
vid.horwitz@uct.ac.za using -f
1312 To: source@collab.sakaiproject.org
1313 From: david.horwitz@uct.ac.za
1325 Author: david.horwitz@uct.ac.za
1335 r39641 | josrodri@iupui.edu | 2007-12-28 23:44:24 +0200 (Fri, 28 Dec
2007)
| 1 line
1340 dhorwitz@david-horwitz-6:~/branchManagemnt/sakai_2-5-x> svn merge -
c39641

https://source.sakaiproject.org/svn/podcasts/trunk podcasts/
1354 From david.horwitz@uct.ac.za Fri Jan 4 04:33:44 2008
1355 Return-Path: <postmaster@collab.sakaiproject.org>
1372 Message-ID: <200801040932.m049W2i5006493@nakamura.uits.iupui.edu>
1377 for <source@collab.sakaiproject.org>;
1381 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 09:33:13 +0000
(GM
T) 1384 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 04:32:03
-0500
1385 Received: (from apache@localhost)
1387 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 04:32:02 -0500
1389 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
da
vid.horwitz@uct.ac.za using -f
1390 To: source@collab.sakaiproject.org
1391 From: david.horwitz@uct.ac.za
1403 Author: david.horwitz@uct.ac.za
1419 r39146 | josrodri@iupui.edu | 2007-12-12 21:40:33 +0200 (Wed, 12 Dec
2007)
| 1 line
1423 dhorwitz@david-horwitz-6:~/branchManagemnt/sakai_2-5-x> svn merge -
c39146
https://source.sakaiproject.org/svn/podcasts/trunk podcasts/
1440 From stephen.marquard@uct.ac.za Fri Jan 4 04:07:34 2008
1441 Return-Path: <postmaster@collab.sakaiproject.org>
1458 Message-ID: <200801040905.m0495rWB006420@nakamura.uits.iupui.edu>
1463 for <source@collab.sakaiproject.org>;
1467 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 09:07:04 +0000
(GM
T) 1470 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 04:05:54
-0500
1471 Received: (from apache@localhost)
1473 for source@collab.sakaiproject.org; Fri, 4 Jan 2008 04:05:53 -0500
1475 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
st
ephen.marquard@uct.ac.za using -f
1476 To: source@collab.sakaiproject.org
1477 From: stephen.marquard@uct.ac.za
1489 Author: stephen.marquard@uct.ac.za
1504 From louis@media.berkeley.edu Thu Jan 3 19:51:21 2008
1505 Return-Path: <postmaster@collab.sakaiproject.org>
1522 Message-ID: <200801040023.m040NpCc005473@nakamura.uits.iupui.edu>
1527 for <source@collab.sakaiproject.org>;
1531 for <source@collab.sakaiproject.org>; Fri, 4 Jan 2008 00:25:00 +0000
(GM
T) 1534 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 19:23:51
-0500
1535 Received: (from apache@localhost)
1537 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 19:23:51 -0500
1539 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
lo
uis@media.berkeley.edu using -f
1540 To: source@collab.sakaiproject.org

1541 From: louis@media.berkeley.edu
1553 Author: louis@media.berkeley.edu
1569 From louis@media.berkeley.edu Thu Jan 3 17:18:23 2008
1570 Return-Path: <postmaster@collab.sakaiproject.org>
1587 Message-ID: <200801032216.m03MGhDa005292@nakamura.uits.iupui.edu>
1592 for <source@collab.sakaiproject.org>;
1596 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 22:17:52 +0000
(GM
T) 1599 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 17:16:43
-0500
1600 Received: (from apache@localhost)
1602 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 17:16:43 -0500
1604 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
lo
uis@media.berkeley.edu using -f
1605 To: source@collab.sakaiproject.org
1606 From: louis@media.berkeley.edu
1618 Author: louis@media.berkeley.edu
1634 From ray@media.berkeley.edu Thu Jan 3 17:07:00 2008
1635 Return-Path: <postmaster@collab.sakaiproject.org>
1652 Message-ID: <200801032205.m03M5Ea7005273@nakamura.uits.iupui.edu>
1657 for <source@collab.sakaiproject.org>;
1661 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 22:06:23 +0000
(GM
T) 1664 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 17:05:14
-0500
1665 Received: (from apache@localhost)
1667 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 17:05:14 -0500
1669 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
ra
y@media.berkeley.edu using -f
1670 To: source@collab.sakaiproject.org
1671 From: ray@media.berkeley.edu
1683 Author: ray@media.berkeley.edu
1698 From cwen@iupui.edu Thu Jan 3 16:34:40 2008
1699 Return-Path: <postmaster@collab.sakaiproject.org>
1716 Message-ID: <200801032133.m03LX3gG005191@nakamura.uits.iupui.edu>
1721 for <source@collab.sakaiproject.org>;
1725 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 21:34:12 +0000
(GM
T) 1728 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 16:33:03
-0500
1729 Received: (from apache@localhost)
1731 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 16:33:03 -0500
1733 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
cw
en@iupui.edu using -f
1734 To: source@collab.sakaiproject.org
1735 From: cwen@iupui.edu
1747 Author: cwen@iupui.edu
1763 From cwen@iupui.edu Thu Jan 3 16:29:07 2008
1764 Return-Path: <postmaster@collab.sakaiproject.org>
1781 Message-ID: <200801032127.m03LRUqH005177@nakamura.uits.iupui.edu>

```

1786 for <source@collab.sakaiproject.org>;
1790 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 21:28:38 +0000
(GM
T) 1793 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 16:27:30
-0500
1794 Received: (from apache@localhost)
1796 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 16:27:30 -0500
1798 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
    cw
en@iupui.edu using -f
1799 To: source@collab.sakaiproject.org
1800 From: cwen@iupui.edu
1812 Author: cwen@iupui.edu
1824 r39403 | wagnermr@iupui.edu | 2007-12-17 17:11:08 -0500 (Mon, 17 Dec
2007)
| 3 lines
1838 From cwen@iupui.edu Thu Jan 3 16:23:48 2008
1839 Return-Path: <postmaster@collab.sakaiproject.org>
1856 Message-ID: <200801032122.m03LMFo4005148@nakamura.uits.iupui.edu>
1861 for <source@collab.sakaiproject.org>;
1865 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 21:23:24 +0000
(GM
T) 1868 for <source@collab.sakaiproject.org>; Thu, 3 Jan 2008 16:22:15
-0500
1869 Received: (from apache@localhost)
1871 for source@collab.sakaiproject.org; Thu, 3 Jan 2008 16:22:15 -0500
1873 X-Authentication-Warning: nakamura.uits.iupui.edu: apache set sender to
    cw
en@iupui.edu using -f
1874 To: source@collab.sakaiproject.org
1875 From: cwen@iupui.edu
1887 Author: cwen@iupui.edu
1899 r35014 | wagnermr@iupui.edu | 2007-09-12 16:17:59 -0400 (Wed, 12 Sep
2007)
| 3 lines

```

```

In [107... # Search for lines that start with 'X' followed by any non
# whitespace characters and ':' #
# followed by a space and any number.
# The number can include a decimal. pattern = r"^X\S+:
\d+(\.\d+)?" #decimal part is optional

for i, line in enumerate(lines, start=1):
    if re.match(pattern, line):
        print(i, line.strip())

```

45 X-DSPAM-Confidence: 0.8475
46 X-DSPAM-Probability: 0.0000
110 X-DSPAM-Confidence: 0.6178
111 X-DSPAM-Probability: 0.0000
175 X-DSPAM-Confidence: 0.6961
176 X-DSPAM-Probability: 0.0000
239 X-DSPAM-Confidence: 0.7565
240 X-DSPAM-Probability: 0.0000
307 X-DSPAM-Confidence: 0.7626
308 X-DSPAM-Probability: 0.0000
380 X-DSPAM-Confidence: 0.7556
381 X-DSPAM-Probability: 0.0000
459 X-DSPAM-Confidence: 0.7002
460 X-DSPAM-Probability: 0.0000
538 X-DSPAM-Confidence: 0.7615
539 X-DSPAM-Probability: 0.0000
617 X-DSPAM-Confidence: 0.7601
618 X-DSPAM-Probability: 0.0000
683 X-DSPAM-Confidence: 0.7605
684 X-DSPAM-Probability: 0.0000
749 X-DSPAM-Confidence: 0.6959
750 X-DSPAM-Probability: 0.0000
814 X-DSPAM-Confidence: 0.7606
815 X-DSPAM-Probability: 0.0000
880 X-DSPAM-Confidence: 0.7559
881 X-DSPAM-Probability: 0.0000
949 X-DSPAM-Confidence: 0.7605
950 X-DSPAM-Probability: 0.0000
1014 X-DSPAM-Confidence: 0.6932
1015 X-DSPAM-Probability: 0.0000
1091 X-DSPAM-Confidence: 0.7558
1092 X-DSPAM-Probability: 0.0000
1165 X-DSPAM-Confidence: 0.6526
1166 X-DSPAM-Probability: 0.0000
1251 X-DSPAM-Confidence: 0.6948
1252 X-DSPAM-Probability: 0.0000
1320 X-DSPAM-Confidence: 0.6528
1321 X-DSPAM-Probability: 0.0000
1398 X-DSPAM-Confidence: 0.7002
1399 X-DSPAM-Probability: 0.0000
1484 X-DSPAM-Confidence: 0.7554
1485 X-DSPAM-Probability: 0.0000
1548 X-DSPAM-Confidence: 0.6956
1549 X-DSPAM-Probability: 0.0000
1613 X-DSPAM-Confidence: 0.6959
1614 X-DSPAM-Probability: 0.0000
1678 X-DSPAM-Confidence: 0.7556
1679 X-DSPAM-Probability: 0.0000
1742 X-DSPAM-Confidence: 0.9846
1743 X-DSPAM-Probability: 0.0000
1807 X-DSPAM-Confidence: 0.8509
1808 X-DSPAM-Probability: 0.0000
1882 X-DSPAM-Confidence: 0.9907

```

# Search for lines that start with 'X' followed by any
# non whitespace characters and ':' followed by a
# space # and any number. The number can include a
# decimal.
# Then print the number if it is greater than zero.
#ur code here
pattern = r"^X\S+: (\d+(\.\d+)?)"

for i, line in enumerate(lines, start=1):
    match = re.match(pattern, line)
    if match:
        number = float(match.group(1))    # convert to
        numeric if number > 0:
            print(i, number)

```

1883 X-DSPAM-Probability: 0.0000 In [108...

```

45 0.8475
110 0.6178
175 0.6961
239 0.7565
307 0.7626
380 0.7556
459 0.7002
538 0.7615
617 0.7601
683 0.7605
749 0.6959
814 0.7606
880 0.7559
949 0.7605
1014 0.6932
1091 0.7558
1165 0.6526
1251 0.6948
1320 0.6528
1398 0.7002
1484 0.7554
1548 0.6956
1613 0.6959
1678 0.7556
1742 0.9846

```

```
1807 0.8509
1882 0.9907
```

```
In [109... # Search for lines that start with 'Details: rev='
# followed by numbers
# Then print the number if one is found
#ur code here pattern =

r"^Details: rev=(\d+)"

for i, line in enumerate(lines, start=1):
    match = re.match(pattern, line)
    if match:
        number = int(match.group(1)) # extract the number if
        number > 0:

        print(i, number)
```

```
In [110... # Search for lines that start with From and a character
# followed by a two digit number between 00 and 99 followed by ':'
# Then print the number if one is found
pattern = r"From.\d{2}:"

for i, line in enumerate(lines, start=1):
    if re.match(pattern, line):
        print(i, line.strip())
```

```
In [111... #Escape character
#ur code here
import string

for i, line in enumerate(lines, start=1):
    # sec character in line is non-printable (excluding newline)
    if any(c not in string.printable for c in line):
        print(i, repr(line.strip()))
```

Task B: Apply Regular Expressions to Perform Text Preprocessing

Use Python and regular expressions (`re` module) to perform the following tasks on a given corpus of text:

i. Text Normalization

Standardize the text by:

- Converting all characters to lowercase.
- Removing all special characters (retain only alphabets and spaces).

ii. HTML Tag Removal

Given a web-scraped text containing HTML content, write a regular expression to remove all HTML tags (e.g., `<p>` , `<div>` , etc.).

iii. Sentence Tokenization

Split a corpus of text into individual sentences.

- Consider punctuation marks like . , ! , and ? as sentence boundaries.
- Ensure the result is a list of clean sentence strings.

iv. Word Tokenization

Split the text into individual words.

- Handle punctuation marks appropriately (e.g., remove them or separate them from words).
- Normalize whitespace (e.g., tabs, multiple spaces, newlines) into single spaces before splitting.

Instructions:

- For each sub-task, define a Python function using regular expressions to accomplish the task.
- Test your function on a sample string.
- Print both the input and the output to demonstrate the transformation.

Example modules you may use: `import re`

In [112... `sample_text = """`

```
<p>Yoooooooo! My names shwen.</p>
Howsssss everyone      doingggggg.
Can we clean this properly? Let's
see!
"""
```

In [113...

```
#NORMALIZATION def
normalize_text(text): #
    Convert to lowercase text
    = text.lower()
    # Remove all non-alphabetic characters (keep spaces)
    text = re.sub(r'^a-z\s]', '', text) #
    Normalize multiple spaces to single space
    text = re.sub(r'\s+', ' ', text).strip()
    return text

print("Input:", sample_text)
print("Output:", normalize_text(sample_text))
```

Input:

```
<p>Yoooooooo! My names shwen.</p>
Howsssss everyone      doingggggg.
Can we clean this properly? Let's see!
```

Output: pyoooooooo my names shwenp howsssss everyone doingggggg can we clean this
p roperly lets see

In [114... `#remove html tags`

```
def remove_html_tags(text): return
re.sub(r'<.*?>', '', text)
print("Input:", sample_text)

print("Output:", remove_html_tags(sample_text))
```

Input:
<p>Yoooooooo! My names shwen.</p>
Howsssss everyone doingggggg.
Can we clean this properly? Let's see!

Output:
Yoooooooo! My names shwen.
Howsssss everyone doingggggg.
Can we clean this properly? Let's see!

In [115... `#sentence tokenization def`

```
sentence_tokenize(text):
    sentences = re.split(r'(?=[.!?])\s+', text) # split text at ., !, ?
    follo return [s.strip() for s in sentences if s]

print("Input:", sample_text)
print("Output:", sentence_tokenize(remove_html_tags(sample_text)))
```

Input:
<p>Yoooooooo! My names shwen.</p>
Howsssss everyone doingggggg.
Can we clean this properly? Let's see!

Output: ['Yoooooooo!', 'My names shwen.', 'Howsssss everyone doingggggg.',
'Can we clean this properly?', 'Let's see!']

In [116... `#word tokenization def`

```
word_tokenize(text):
    text = re.sub(r'[^w\s]', '', text) # punctuation
    text = re.sub(r'\s+', ' ', text).strip() #
    whitespace return text.split() # Split into words

print("Input:", sample_text)
print("Output:", word_tokenize(remove_html_tags(sample_text)))
```

Input:
<p>Yoooooooo! My names shwen.</p>
Howsssss everyone doingggggg.
Can we clean this properly? Let's see!

Output: ['Yoooooooo', 'My', 'names', 'shwen', 'Howsssss', 'everyone',
'doinggagg',
'Can', 'we', 'clean', 'this', 'properly', 'Lets', 'see']

